

SUMMER EXAMINATION 2026
Natural Language Processing
BSIT - 3rd Semester Time: 90 Minutes Max Marks: 80

Name: _____ Roll No.: _____ Date: _____

1. NLP concerns computational processing of
(A) human language (B) electrical circuits
(C) database pages only (D) image pixels only
2. Language is difficult because meaning depends on
(A) context and ambiguity (B) font size
(C) file extension (D) network speed only
3. The smallest meaningful linguistic unit is a
(A) morpheme (B) document
(C) paragraph (D) parse forest only
4. A free morpheme can
(A) stand alone as a word (B) never occur alone
(C) only be a suffix (D) only be punctuation
5. Lemmatization reduces word forms to a
(A) dictionary base form (B) random stem
(C) character code (D) sentence label
6. Stemming usually removes affixes using
(A) heuristic rules (B) a full world model
(C) audio filters (D) database joins
7. Tokenization splits text into
(A) tokens (B) class weights
(C) parse probabilities only (D) image regions
8. A stop word is often removed because it is
(A) very frequent and less informative (B) always misspelled
(C) a named entity (D) a rare technical term
9. Lowercasing can reduce vocabulary but may lose
(A) case information (B) token order
(C) all punctuation (D) sentence boundaries only
10. A regular expression is a pattern for matching
(A) text strings (B) neural weights
(C) database rows (D) audio frequencies only
11. In regex, a period commonly matches
(A) a single character (B) a digit only
(C) the end of input only (D) a word boundary only
12. An FSA recognizes regular languages using states and
(A) transitions (B) gradients
(C) attention heads (D) parse probabilities
13. POS tagging assigns each token a
(A) syntactic category (B) vector norm
(C) document ID (D) sentiment score only
14. A noun phrase commonly contains a
(A) noun head (B) verb root only
(C) transition matrix (D) discount factor
15. A context-free grammar uses productions with a
(A) single nonterminal on the left (B) probability only
(C) vector on the left (D) regular expression only
16. A parse tree represents
(A) hierarchical syntactic structure (B) document frequency
(C) audio amplitude (D) word embeddings only
17. Constituency parsing groups tokens into
(A) phrases or constituents (B) database tables
(C) attention scores only (D) classes
18. Dependency parsing represents head-dependent
(A) relations (B) pixel neighborhoods
(C) document counts (D) reward paths
19. Ambiguity occurs when a sentence has
(A) multiple plausible analyses (B) no tokens
(C) only one word (D) no grammar
20. An n-gram model predicts a token from
(A) the previous n-1 tokens (B) the next document only
(C) image pixels (D) a parse tree only
21. The Markov assumption limits dependence to
(A) a fixed history (B) all future tokens
(C) the full corpus always (D) no context
22. Bigram probability models depend on
(A) one previous token (B) two previous tokens exactly
(C) the whole paragraph (D) only the label
23. Smoothing assigns some probability to
(A) unseen events (B) only frequent words
(C) punctuation only (D) parse roots
24. Add-one smoothing adds
(A) one to counts (B) one token to every sentence
(C) one class (D) one hidden layer
25. Perplexity measures how well a language model
(A) predicts a sequence (B) tags images
(C) retrieves documents only (D) detects entities only
26. Lower perplexity generally indicates
(A) better predictive performance (B) worse tokenization always
(C) higher ambiguity (D) fewer vocabulary items
27. One-hot encoding represents each word as
(A) a sparse indicator vector (B) a probability table
(C) a parse tree (D) a dense audio signal
28. Bag-of-words ignores
(A) word order (B) word identity
(C) document membership (D) term counts
29. TF measures a term's frequency in a
(A) document (B) corpus only
(C) query class (D) parse tree
30. IDF downweights terms that occur in
(A) many documents (B) one document only
(C) no documents (D) one sentence only
31. TF-IDF is high for terms frequent in a document but
(A) rare in the corpus (B) common everywhere
(C) absent from the document (D) only punctuation
32. Word embeddings represent words as
(A) dense vectors (B) binary labels only
(C) regular expressions (D) parse rules only
33. Distributional semantics is based on words occurring in similar
(A) contexts (B) fonts
(C) file names (D) audio channels
34. Cosine similarity measures the angle-based similarity between
(A) vectors (B) parse trees only
(C) documents by length only (D) probabilities only
35. The cosine similarity of identical nonzero vectors is
(A) 1 (B) 0
(C) -1 (D) undefined always
36. An inverted index maps terms to lists of
(A) documents or positions (B) POS tags only
(C) embedding dimensions (D) audio frames
37. The vector space model represents documents as
(A) vectors of term weights (B) parse graphs only
(C) database schemas (D) class labels
38. Precision in retrieval is relevant results divided by
(A) retrieved results (B) all relevant results
(C) all documents (D) false negatives
39. Recall in retrieval is relevant results divided by
(A) all relevant results (B) retrieved results
(C) all queries (D) true negatives
40. If 6 of 10 retrieved documents are relevant, precision is
(A) 0.6 (B) 0.4
(C) 1.6 (D) 6.0

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1. Word sense disambiguation selects the intended
(A) meaning of a word (B) sentence length
(C) POS tag only (D) document ID
2. NER identifies spans such as people, places, and
(A) organizations (B) grammars
(C) vectors (D) n-grams only
3. Semantic role labeling identifies predicates and their
(A) arguments and roles (B) characters only
(C) document frequencies (D) attention heads only
4. Coreference resolution links expressions that refer to the same
(A) entity (B) token count
(C) parse rule (D) audio frame
5. An HMM contains hidden states and
(A) observations (B) attention layers
(C) database indexes (D) word vectors only
6. In an HMM, POS tags can be hidden states and words can be
(A) observations (B) transitions
(C) rewards (D) documents
7. The Viterbi algorithm finds the most likely
(A) state sequence (B) document ranking
(C) embedding (D) parse grammar only
8. The forward algorithm computes total probability using
(A) dynamic programming (B) greedy stemming
(C) TF-IDF (D) beam width only
9. RNNs model sequences using a recurrent
(A) hidden state (B) inverted index
(C) parse node (D) vocabulary hash
10. LSTMs address vanishing gradients using
(A) gates and cell state (B) one-hot vectors
(C) stop words (D) IDF
11. GRUs use update and reset
(A) gates (B) parse trees
(C) document terms (D) speech codecs only
12. Seq2seq models map an input sequence to
(A) an output sequence (B) one scalar only
(C) a database row (D) a fixed image
13. Attention lets a decoder focus on
(A) relevant source representations (B) all tokens equally always
(C) only the final token (D) training labels only
14. Self-attention relates tokens within
(A) the same sequence (B) different databases
(C) only audio files (D) a single class label
15. In scaled dot-product attention, scores use queries and
(A) keys (B) labels
(C) stems (D) documents only
16. The value vectors in attention carry
(A) content to aggregate (B) class names
(C) grammar rules (D) document IDs only
17. Multi-head attention learns
(A) multiple representation subspaces (B) multiple databases
(C) one token only (D) a single scalar
18. A transformer avoids recurrence by using
(A) attention and positional information (B) only convolutions
(C) only HMMs (D) no token order
19. Positional encoding supplies information about
(A) token order (B) word frequency only
(C) document labels (D) audio pitch only
20. BERT is primarily an
(A) encoder-style bidirectional model (B) decoder-only model
(C) HMM (D) finite-state lexer
21. GPT is primarily a
(A) decoder-only autoregressive model (B) bidirectional en-
coder only
(C) database engine (D) speech recognizer only
22. Masked language modeling hides tokens and predicts the
(A) masked tokens (B) next document
(C) POS dictionary (D) audio waveform only
23. Autoregressive generation predicts tokens from
(A) previous context (B) future gold tokens
(C) random classes (D) image regions only
24. Sentiment analysis classifies
(A) opinion polarity (B) syntax trees only
(C) document length (D) speaker identity only
25. Machine translation converts text between
(A) languages (B) formats only
(C) databases (D) audio channels only
26. ASR converts speech audio to
(A) text (B) text to images
(C) tokens to audio only (D) labels to vectors
27. TTS converts text into
(A) speech audio (B) images
(C) parse trees (D) database rows
28. BLEU evaluates translation using
(A) n-gram overlap (B) entity spans only
(C) word vectors only (D) audio quality only
29. ROUGE is commonly used for
(A) summarization overlap (B) POS tagging
(C) speech synthesis (D) database retrieval only
30. Classification accuracy is correct predictions divided by
(A) all predictions (B) predicted positives only
(C) actual positives only (D) retrieved documents
31. A language model's log probability is useful because products of probabilities can
(A) underflow numerically (B) increase vocabulary
(C) remove syntax (D) create labels
32. The log of a probability between 0 and 1 is
(A) nonpositive (B) always positive
(C) always zero (D) undefined
33. A dataset can contain bias when its language or labels
(A) underrepresent groups (B) are tokenized
(C) use vectors (D) have punctuation
34. An NLP system should protect
(A) private and sensitive text (B) only stop words
(C) only common nouns (D) only punctuation
35. A 3-word sentence has how many contiguous bigrams?
(A) 2 (B) 3
(C) 1 (D) 4
36. If a term appears in 2 of 10 documents, its document frequency is
(A) 2 (B) 5
(C) 8 (D) 10
37. A query with 4 relevant retrieved results and 2 irrelevant results has precision
(A) 4/6 (B) 2/4
(C) 4/10 (D) 6/4
38. A beam search keeps a fixed number of
(A) best partial hypotheses (B) training documents
(C) POS tags (D) attention matrices only
39. Character n-grams can help handle
(A) unknown word forms (B) database transactions
(C) image rotations (D) reward discounting
40. In parsing, a terminal symbol is typically a
(A) word or token (B) grammar variable
(C) document ID (D) hidden state only